

HIGH-RESOLUTION SPECTROSCOPIC OBSERVATIONS OF A NEW CH SUBGIANT STAR: BD–03°3668

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Received 2009 December 22; accepted 2010 December 14; published 2011 February 2

ABSTRACT

This work reports on the discovery of a new CH subgiant star BD–03°3668 with a significant overabundance of the elements created by *s*-process nucleosynthesis, such as barium and lead. Its atmospheric parameters and abundance pattern are determined by employing the local thermodynamic equilibrium model atmospheres of Kurucz using the spectral analysis code MOOG. Due to the low luminosity of this star, the *s*-process overabundance is better explained by mass transfer from an asymptotic giant branch star in the past.

Key words: stars: abundances – stars: chemically peculiar – stars: individual (BD–03°3668)

1. INTRODUCTION

Barium and subgiant CH stars are chemically peculiar stars displaying both anomalously high heavy-element ($Z > 30$) abundances and overabundance of carbon compared to the Sun, with $[C/Fe]$ varying from +0.4 to +1.2 (Snedden & Bond 1976; Luck & Bond 1982, 1991; Smith 1984; Smith et al. 1993; North et al. 1994; Zács 1994; Antipova et al. 2003, 2004; Pereira & Junqueira 2003; Allen & Barbuy 2006; Drake & Pereira 2008; Pereira & Drake 2009). In the mass-transfer hypothesis, the observed chemical peculiarities of these stars are a consequence of mass transfer through stellar winds in a binary system from an asymptotic giant branch (AGB) star (now a white dwarf) to a less evolved companion, a barium giant or a subgiant CH star. Therefore, the study of their chemical peculiarities is important for understanding the physics of accretion phenomena by comparing the number of known barium and subgiant CH stars with their theoretical birthrate (Han et al. 1995). In addition, this may provide a better understanding of the *s*-process nucleosynthesis operating in AGB stars.

Busso et al. (2001) use the $[hs/ls]^3$ index for several classes of chemically peculiar stars as a criterion to measure the amount of their enrichment through the *s*-process in order to test whether current AGB models account for the observed abundance distributions. Likewise, a high-resolution spectroscopy survey dedicated to analyzing and measuring the abundance pattern of a large sample of a uniform population of chemically peculiar stars would also set another constraint on those models. Barium giants could be these targets. They make up the largest known sample among those kinds of stars, even larger than that of the CEMP *s*-rich ones (Snedden et al. 2008) which are probably binaries. The determination of the abundance pattern of the elements created by *s*-process nucleosynthesis, as well as the carbon abundance and the $^{12}C/^{13}C$ isotopic ratio, would be very useful not only in investigations of some of the issues raised, but also in investigating mixing processes, such as thermohaline mixing, for example (Stancliffe et al. 2007; Denissenkov & Pinsonneault 2008; Pereira & Drake 2009). Such surveys could also be used to probe different populations of barium stars by measuring their metallicities, for instance, by improving the number of known barium stars in the halo (Drake & Pereira

2008; Pereira & Drake 2009). Motivated by the questions mentioned above, we started a high-resolution spectroscopic survey of the barium stars from the MacConnell et al. (1972) and Bidelman (1981) samples. Among about 230 stars surveyed, we discovered a new CH subgiant, BD–03°3668 or MFU 99 (star no. 99 of MacConnell et al. 1972) which is the subject of the present work.

Subgiant CH stars are very rare objects. They have been found in studies using objective prism surveys (Bond 1974) or in extensive surveys. Gómez et al. (1997) used the maximum-likelihood method to analyze a sample of 300 barium stars taken from Lu (1991) and were able to separate the stars into groups, according to their absolute magnitude and kinematics. Sixteen CH subgiant candidates were found, 5% of the sample. Despite the fact that not all of them proved to be CH subgiants (Allen & Barbuy 2006), some were later spectroscopically confirmed to belong to the group of the new CH subgiant stars, HD 8270, HD 13551, and HD 22589 (Pereira 2005). Spectroscopic abundance surveys of Galactic F and G dwarfs were also able to reveal new CH subgiant stars (HR 107, Tomkin et al. 1989; HD 6434, HR 2906, HR 4285, HR 4395, and HR 5338, Edvardsson et al. 1993; HD 21922, HD 36667, HD 80218, HD 140324 and HD 220842, Reddy et al. 2003). These newly discovered CH subgiants made up only about 3% of the total sample. More recently Gray & Griffin (2007) identified a new barium dwarf star, HD 26367, through the NSTARS Project. According to Han et al. (1995), the known number of CH subgiants seems to be in contradiction with theoretical predictions. Therefore, finding and confirming a new CH subgiant can help to constrain the expected birthrate of these subgiants.

2. OBSERVATIONS

The high-resolution spectrum of BD–03°3668 analyzed in this work was obtained at the 2.2 m telescope of La Silla (Chile) on 2008 April 8, with the FEROS (Fiberfed Extended Range Optical Spectrograph) echelle spectrograph (Kaufer et al. 1999). Its spectral resolving power is $R = 48,000$ corresponding to 2.2 pixels of $15 \mu m$ and the wavelength coverage is from 3800 Å to 9200 Å. The nominal signal-to-noise ratio S/N was evaluated by the measurement of the rms flux fluctuation in selected continuum windows, and a typical achieved value was $S/N = 150$ –200, after 1500 s of integration time.

³ $[hs/ls] = \log (hs/ls)_* - \log (hs/ls)_\odot$ where $[hs]$ and $[ls]$ are the mean abundances of the *s*-elements at the Ba and Zr peaks, respectively.

Table 1
Observed Fe I and Fe II Lines

Element	λ (Å)	χ (eV)	$\log gf$	W_λ (mÅ)
Fe I	5159.06	4.28	-0.650	52
	5194.94	1.56	-2.090	143
	5198.71	2.22	-2.140	104
	5242.49	3.63	-0.970	78
	5281.79	3.04	-0.830	126
	5288.52	3.69	-1.510	46
	5315.05	4.37	-1.400	26
	5321.11	4.43	-1.190	48
	5322.04	2.28	-2.840	60
	5339.93	3.27	-0.680	124
	5373.71	4.47	-0.710	56
	5389.48	4.42	-0.250	74
	5393.17	3.24	-0.720	111
	5400.50	4.37	-0.100	106
	5417.03	4.42	-1.530	32
	5445.04	4.39	0.040	96
	5506.78	0.99	-2.800	129
	5522.45	4.21	-1.400	38
	5532.75	3.57	-2.000	50
	5554.90	4.55	-0.380	75
	5560.21	4.43	-1.040	38
	5563.60	4.19	-0.840	59
	5569.61	3.42	-0.499	133
	5576.09	3.43	-0.849	110
	5638.26	4.22	-0.720	69
	5686.53	4.55	-0.457	70
	5691.50	4.30	-1.370	38
	5705.46	4.30	-1.360	30
	5717.83	4.28	-0.979	52
	5731.76	4.26	-1.150	55
	5762.99	4.21	-0.410	95
	5806.73	4.61	-0.900	38
	5814.81	4.28	-1.820	15
	5852.22	4.55	-1.180	33
	5883.82	3.96	-1.209	56
	5916.25	2.45	-2.990	55
	5934.65	3.93	-1.020	77
	6016.66	3.54	-1.669	71
	6024.06	4.55	-0.060	90
	6027.05	4.08	-1.090	56
	6056.01	4.73	-0.400	57
	6065.48	2.61	-1.530	115
	6079.01	4.65	-0.969	31
	6082.71	2.22	-3.580	35
	6093.64	4.61	-1.350	23
	6096.66	3.98	-1.780	26
	6136.61	2.45	-1.400	124
	6137.69	2.59	-1.400	127
	6151.62	2.18	-3.290	52
	6170.51	4.80	-0.380	70
	6173.34	2.22	-2.880	70
	6187.99	3.94	-1.570	40
	6191.56	2.43	-1.420	135
	6200.32	2.60	-2.440	72
	6213.43	2.22	-2.480	84
	6252.56	2.40	-1.720	112
	6265.13	2.18	-2.550	91
	6322.69	2.59	-2.430	77
	6380.74	4.19	-1.320	46
	6393.60	2.43	-1.430	126
	6411.65	3.65	-0.660	109
	6419.95	4.73	-0.090	75
	6421.35	2.28	-2.010	112
	6430.85	2.18	-2.010	114
	6469.19	4.84	-0.619	47
	6592.91	2.72	-1.470	104

Table 1
(Continued)

Element	λ (Å)	χ (eV)	$\log gf$	W_λ (mÅ)
Fe I	6593.87	2.44	-2.420	80
	6597.56	4.80	-0.920	31
	6609.11	2.56	-2.690	63
	6750.15	2.42	-2.619	76
	6752.71	4.64	-1.200	22
	6810.26	4.61	-1.200	37
	6820.37	4.64	-1.170	29
	6858.15	4.61	-0.930	41
	7130.92	4.22	-0.700	86
	7132.99	4.08	-1.609	32
	4993.35	2.81	-3.670	39
	5197.56	3.23	-2.250	71
	5234.62	3.22	-2.240	66
	5276.00	3.20	-1.910	74
	5284.10	2.89	-3.010	45
	5325.56	3.22	-3.170	23
	5425.25	3.20	-3.210	40
	5534.83	3.25	-2.770	47
	5991.37	3.15	-3.560	20
	6149.25	3.89	-2.720	20
Fe II	6247.55	3.89	-2.340	35
	6416.92	3.89	-2.680	31
Fe II	6432.68	2.89	-3.580	31

3. ANALYSIS AND RESULTS

3.1. Line Selection, Measurement, and Oscillator Strengths

The spectrum of BD-03°3668 shows many atomic Fe I and Fe II absorption lines as well as transitions due to Na I, Mg I, Al I, Si I, Ca I, Ti I, Cr I, Mn I, Ni I, Cu I, Zn I, Y II, Zr I, Ba II, La II, Ce II, Nd II, and Pb I. The chosen lines are sufficiently unblended to yield reliable abundances. They are listed in Table 1 for Fe I and Fe II and in Table 2 for other elements. Table 1 shows the Fe I and Fe II lines employed in the analysis, the lower excitation potentials χ (eV) of the transitions, the gf -values, and the measured equivalent widths. These were obtained by fitting Gaussian profiles to the observed absorption lines. The gf -values for the Fe I and Fe II lines in Table 1 were taken from Lambert et al. (1996). The sources of the gf -values for the other elements are given in Table 2.

3.2. Determination of the Atmospheric Parameters

The determination of the stellar atmospheric parameters, effective temperature (T_{eff}), surface gravity ($\log g$), microturbulence (ξ), and metallicity ($[\text{Fe}/\text{H}]$) (where $[\text{X}/\text{H}] = \log(N(\text{X})/N(\text{H}))_\star - \log(N(\text{X})/N(\text{H}))_\odot$) are prerequisites to finding the photospheric abundances. The method has already been described in previous papers dedicated to studies of barium giants and dwarfs (Drake & Pereira 2008; Pereira & Junqueira 2003; Pereira 2005). In brief, the gravity was determined by forcing the Fe I and Fe II lines to yield the same iron abundance at the selected effective temperature. The microturbulent velocity was obtained by requiring that the abundance, determined from individual Fe I lines, shows no dependence on the equivalent width. The solution for the excitation equilibrium was set to the zero slope of the trend between the Fe I abundances and the excitation potentials of the measured lines. The atmospheric parameters were determined employing the local thermodynamic equilibrium (LTE) model atmospheres of Kurucz (1993) using a recent version

Table 2
Other Lines Studied

λ (Å)	Element	χ (eV)	$\log gf$	Ref	W_λ (mÅ)
5682.65	Na I	2.10	-0.700	PS	91
5688.22	Na I	2.10	-0.400	PS	110
6698.67	Al I	3.14	-1.630	R03	27
7835.32	Al I	4.04	-0.580	R03	42
7836.13	Al I	4.02	-0.400	R03	46
8772.88	Al I	4.02	-0.250	R03	48
8773.91	Al I	4.02	-0.070	R03	78
8712.69	Mg I	5.93	-1.260	WSM	66
8717.83	Mg I	5.91	-0.709	WSM	75
8736.04	Mg I	5.94	-0.339	WSM	103
5793.08	Si I	4.93	-2.066	R03	38
6125.03	Si I	5.61	-1.540	E93	33
6131.58	Si I	5.62	-1.685	E93	24
6145.08	Si I	5.62	-1.480	E93	30
8728.01	Si I	6.18	-0.360	E93	61
8742.45	Si I	5.87	-0.510	E93	93
6102.73	Ca I	1.88	-0.789	D2002	131
6161.30	Ca I	2.52	-1.270	E93	80
6166.44	Ca I	2.52	-1.140	R03	76
6169.04	Ca I	2.52	-0.800	R03	91
6169.56	Ca I	2.53	-0.480	DS91	108
6455.60	Ca I	2.51	-1.290	R03	64
6471.66	Ca I	2.51	-0.690	S86	94
6493.79	Ca I	2.52	-0.110	DS91	123
4533.25	Ti I	0.85	0.530	D2002	126
5866.46	Ti I	1.07	-0.871	E93	72
6091.18	Ti I	2.27	-0.370	R03	21
6261.10	Ti I	1.43	-0.480	B86	68
5296.70	Cr I	0.98	-1.390	GS88	103
5300.75	Cr I	0.98	-2.130	GS88	66
5345.81	Cr I	1.00	-0.980	GS88	136
5348.33	Cr I	1.00	-1.290	GS88	107
5409.80	Cr I	1.03	-0.720	GS88	92
6330.10	Cr I	0.94	-2.920	R03	41
5084.10	Ni I	3.68	0.060	E93	73
5115.40	Ni I	3.83	-0.280	R03	58
6176.82	Ni I	4.09	-0.264	R03	50
6327.60	Ni I	1.68	-3.113	MFW	51
6482.80	Ni I	1.94	-2.630	MFW	45
6586.33	Ni I	1.95	-2.810	MFW	37
6643.64	Ni I	1.68	-2.030	MFW	105
6772.32	Ni I	3.66	-0.970	R03	44
7788.93	Ni I	1.95	-1.990	E93	97
4810.53	Zn I	4.06	-0.170	BG80	64
4883.68	Y II	1.08	0.070	SN96	114
5087.43	Y II	1.08	-0.170	SN96	91
5200.41	Y II	0.99	-0.570	SN96	81
5205.72	Y II	1.03	-0.340	SN96	113
5289.81	Y II	1.03	-1.850	R03	36
5402.78	Y II	1.84	-0.440	R03	52
6127.46	Zr I	0.15	-1.060	S96	38
6134.57	Zr I	0.00	-1.280	S96	23
6140.46	Zr I	0.52	-1.410	S96	15
6143.18	Zr I	0.07	-1.100	S96	32
6496.90	Ba II	0.60	-0.379	WM80	262
4934.83	La II	1.25	-0.924	VWR	21
5303.53	La II	0.32	-1.349	VWR	44
5880.63	La II	0.24	-1.830	R03	28
6320.42	La II	0.17	-1.562	VWR	61
6390.48	La II	0.32	-1.561	SN96	54
6774.33	La II	0.12	-1.750	SN96	58
4562.37	Ce II	0.48	0.330	SN96	82
4628.16	Ce II	0.52	0.260	VWR	91
5117.17	Ce II	1.40	0.010	VWR	38
5187.45	Ce II	1.21	0.300	VWR	51
5274.24	Ce II	1.28	0.389	VWR	54

Table 2
(Continued)

λ (Å)	Element	χ (eV)	$\log gf$	Ref	W_λ (mÅ)
5472.30	Ce II	1.25	-0.190	VWR	40
6051.80	Ce II	0.23	-1.600	SN96	28
4811.34	Nd II	0.06	-1.015	VWR	66
4959.12	Nd II	0.06	-0.916	VWR	81
5212.36	Nd II	0.20	-0.700	SN96	71
5234.19	Nd II	0.55	-0.460	SN96	70
5249.58	Nd II	0.98	0.080	SN96	72
5293.16	Nd II	0.82	-0.200	SN96	72
5311.46	Nd II	0.98	-0.560	SN96	54
5319.81	Nd II	0.55	-0.350	SN96	75
5416.38	Nd II	0.86	-0.980	VWR	27
5431.54	Nd II	1.12	-0.457	VWR	40
5442.26	Nd II	0.68	-0.900	SN96	46
5740.88	Nd II	1.16	-0.560	VWR	31

References. BG80: Biemont & Godefroid 1980; D2002: Depagne et al. 2002; DS91: Drake & Smith 1991; E93: Edvardsson et al. 1993; GS88: Gratton & Sneden 1988; MFW: Martin et al. 1988; R03: Reddy et al. 2003; PS: Preston & Sneden 2001; S86: Smith et al. 1986; S96: Smith et al. 1996; SN96: Sneden et al. 1996; VWR: van Winckel & Reyniers 2000; WM80: Wiese & Martin 1980; WSM: Wiese et al. 1969.

Table 3
Atmospheric Parameters of BD-03°3668

Parameter	Value
T_{eff} (K)	$5\,220 \pm 120$
$\log g$ (dex)	3.7 ± 0.3
[Fe/H] (dex)	-0.55 ± 0.13
ξ (km s ⁻¹)	0.9 ± 0.4
V_{rad} (km s ⁻¹)	-56.7 ± 0.5
Gal. latitude (b°)	$+47.9$

of the spectral analysis code MOOG (Sneden 1973). The final adopted atmospheric parameters are given in Table 3.

The internal errors in the adopted effective temperature (T_{eff}) and microturbulent velocity (ξ) can be found from the uncertainty in the slope of the Fe I abundance versus excitation potential and versus reduced equivalent width (W_λ/λ). These quantities are given in Table 3. The standard deviation in $\log g$ was found by varying this parameter around the adopted solution until the difference between Fe I and Fe II mean abundances differed by exactly one standard deviation of the [Fe I/H] mean value.

3.3. Abundance Analysis

In order to obtain the abundances of chemical elements in the photosphere of BD-03°3668, the equivalent widths were calculated through integration using a model atmosphere and comparing the obtained values to the observed ones. The calculation was repeated, changing the abundance of the element in turn, until a match was achieved. The results are given in Table 4, in which n denotes the number of the lines employed for each species and σ the standard deviation of the obtained abundance. To carry out the calculations, the line-synthesis code MOOG was used.

The abundance of manganese is based on the lines of Mn I $\lambda 6021$ and $\lambda 6013$ and the abundance of copper on the Cu I line $\lambda 5782$. The abundances for these elements were obtained through the spectrum synthesis technique. The synthetic spectra

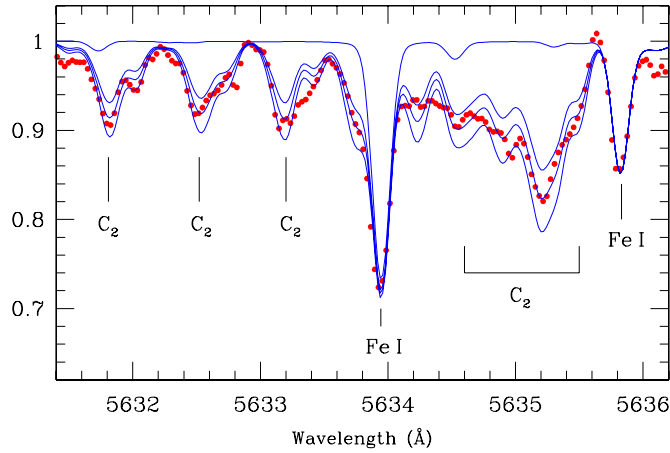


Figure 1. Observed (dotted red line) and synthetic (solid blue line) spectra in the region around the C_2 molecule lines at 5635 Å. Synthetic spectra were computed for four different carbon abundances: $\log \varepsilon(C) = \text{none}, 8.25, 8.35, \text{ and } 8.45$.

Table 4

Abundances in the Scale $\log \varepsilon(H) = 12.0$ and in the Notation $[X/H]$ and $[X/Fe]$

Species	n	$\log \varepsilon(X)$	$[X/H]$	$[X/Fe]$
Fe I	76	6.97 ± 0.12	-0.55	...
Fe II	13	6.97 ± 0.16	-0.55	...
C	2	8.60 ± 0.05	+0.04	+0.59
N	1	7.62 ± 0.20	-0.43	+0.12
O I	4	8.73 ± 0.13	-0.20	+0.35
Na I	2	5.84	-0.49	+0.06
Mg I	3	7.36 ± 0.29	-0.22	+0.33
Al I	5	6.05 ± 0.11	-0.42	+0.13
Si I	6	7.37 ± 0.14	-0.18	+0.37
Ca I	8	6.07 ± 0.16	-0.29	+0.26
Ti I	4	4.78 ± 0.18	-0.24	+0.31
Cr I	6	5.39 ± 0.08	-0.28	+0.27
Mn I	2	4.69	-0.70	-0.15
Ni I	9	5.83 ± 0.16	-0.42	+0.13
Cu I	1	3.81	-0.40	+0.15
Zn I	1	4.18	-0.42	+0.13
Y II	6	2.99 ± 0.18	+0.75	+1.30
Zr I	4	3.50 ± 0.21	+0.90	+1.45
Ba II	1	3.22	+1.09	+1.64
La II	6	2.41 ± 0.24	+1.24	+1.79
Ce II	7	2.64 ± 0.15	+1.18	+1.73
Nd II	12	2.93 ± 0.23	+1.43	+1.98
Pb I	1	3.20	+1.25	+1.80

were broadened to fit the instrumental profile. All hyperfine-structure components for the lines of Cu I and Mn I were explicitly included. The gf -values for these lines were taken from Cunha et al. (2002) and del Peloso et al. (2005), respectively. Table 4 also shows the abundances obtained for these elements. Likewise, the abundances of the light elements, carbon, nitrogen, and oxygen, and the $^{12}C/^{13}C$ isotopic ratio were determined by means of the spectrum synthesis technique employing the LTE atmosphere models of Kurucz (1993). Since in the atmospheres of cool stars the abundances of the CNO elements are interdependent due to the association of carbon and oxygen in the molecules of CO, we carried out an iterative procedure of CNO abundance determinations, until the abundances of these three elements agreed.

The spectral regions used to determine the carbon, nitrogen, and oxygen were the same as used in our last study of the chemically peculiar metal-poor stars HD 10613 and BD+04°2466

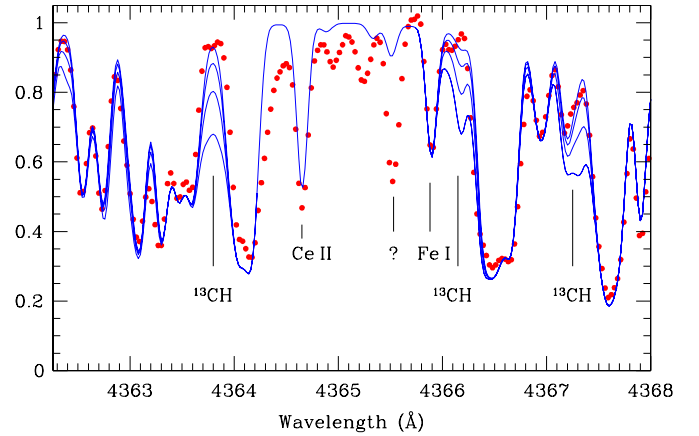


Figure 2. Observed (dotted red line) and synthetic (solid blue line) spectra in the region containing the CH molecular lines at 4365 Å. In the synthetic spectra (from top to bottom), we show the synthesis for the $^{12}C/^{13}C$ ratio = 64, 32, 16, and 8.

(Pereira & Drake 2009). For carbon we used the spectral regions at the C_2 (0, 1) band head of the Swan system $A^3\Pi_g - X^3\Pi_u$ at 5635 Å, the C_2 feature at 5086 Å, and the CH lines of the $A^2\Delta - X^2\Pi$ system at ~ 4365 Å. For nitrogen we made use of a spectral region containing the ^{12}CN lines of the (2, 0) band of the CN red system $A^2\Pi - X^2\Sigma$ in the 7994–8020 Å wavelength range. For oxygen we used the [O I] forbidden line at 6300.304 Å. The observed and synthetic spectra in the region containing the C_2 molecular lines around 5635 Å are shown in Figure 1.

To find the $^{12}C/^{13}C$ isotopic ratio, we considered the spectral region at 4360–4370 Å, which contains the lines of the molecules ^{12}CH and ^{13}CH . We consider three ^{13}CH features at 4363.8, 4366.1, and 4367.2 Å. The observed and synthetic spectra in the region around 4365 Å are shown in Figure 2. The derived carbon isotopic ratio is $^{12}C/^{13}C \geq 32$.

We also determined the lead abundance for BD–03°3668 from the Pb I line at 4057.81 Å. The line data were taken from Van Eck et al. (2003) which include isotopic shifts and hyperfine splitting. The observed and synthetic spectra in the region around the Pb I $\lambda 4058$ line are shown in Figure 3.

The derived abundances of chemical elements in the photosphere of BD–03°3668 are presented in Table 4.

3.4. Abundance Uncertainties

The uncertainties in the derived abundances are dominated by two main sources: the stellar parameters and the equivalent width measurements. The uncertainties due to errors in the stellar atmospheric parameters were computed by varying these quantities in the interval given by the standard deviations in Table 4 and calculating the changes incurred in the element abundances. The results of these calculations for BD–03°3668 are displayed in Columns 2–5 of Table 5.

Errors in the equivalent widths are set essentially by S/N and spectral resolution. The uncertainties due to equivalent width measurements were computed from an expression given by Cayrel (1988). As applied to our spectrum, having $R \approx 48,000$ and a typical S/N of 150, the expected uncertainties in the equivalent widths are about 2–3 mÅ. These error estimates have been applied to the stellar W_λ and the corresponding uncertainties in the element abundances are listed in Column 6 of Table 5.

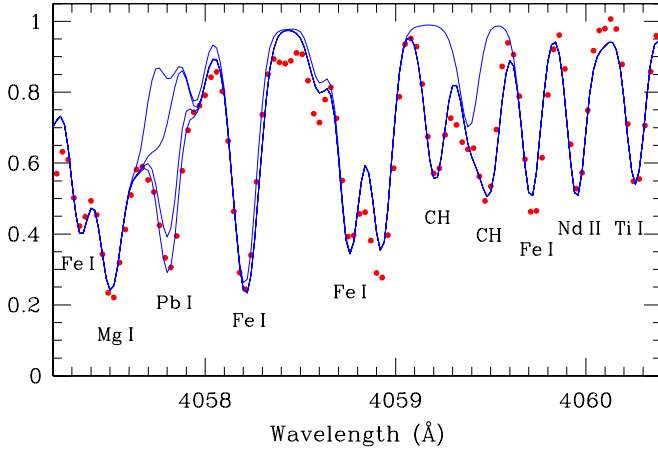


Figure 3. Observed (dotted red line) and synthetic (solid blue line) spectra in the region around the Pb I line at 4057.8 Å. In the synthetic spectra, we show the synthesis for lead abundances of $\log \varepsilon(\text{Pb}) = 0.1, 0.1, 1.4,$ and 1.9 and for carbon abundances of $\log \varepsilon(\text{C}) = \text{none}, 8.29, 8.29,$ and 8.29 .

Table 5
Abundance Uncertainties of BD-03°3668

Species	ΔT_{eff}	$\Delta \log g$	$\Delta \xi$	$\Delta[\text{Fe}/\text{H}]$	ΔW_{λ}	$(\sum \sigma^2)^{1/2}$	σ_{obs}
Fe I	+0.09	0.00	-0.07	+0.02	+0.06	0.13	0.14
Fe II	-0.06	+0.15	-0.08	-0.03	+0.06	0.19	0.16
Na I	+0.09	-0.05	-0.07	+0.01	+0.06	0.14	...
Al I	+0.05	-0.02	-0.03	0.00	+0.03	0.07	0.11
Mg I	+0.04	-0.02	-0.03	0.00	+0.03	0.06	0.23
Si I	0.00	+0.04	-0.04	-0.01	+0.05	0.08	0.14
Ca I	+0.11	-0.02	-0.11	+0.02	+0.06	0.17	0.15
Ti I	+0.15	+0.02	-0.11	+0.02	+0.04	0.19	0.26
Mn I	+0.11	+0.01	0.00	+0.05	+0.04	0.13	...
Cr I	+0.17	-0.02	-0.14	+0.02	+0.04	0.23	0.07
Ni I	+0.08	+0.04	-0.10	0.00	+0.06	0.15	0.14
Cu I	+0.12	-0.01	-0.09	+0.06	+0.05	0.17	...
Y II	+0.01	+0.12	-0.17	-0.04	-0.03	0.21	0.18
Zr I	+0.18	-0.02	-0.02	+0.10	0.06	0.22	0.21
Ba II	+0.05	+0.02	-0.16	-0.05	0.04	0.18	...
La II	+0.03	+0.14	-0.07	-0.03	0.05	0.15	0.17
Ce II	+0.02	+0.13	-0.11	-0.03	0.05	0.18	0.13
Nd II	+0.03	+0.13	-0.11	-0.03	0.04	0.18	0.15

Notes. The second column gives the variation of the abundances caused by +120 K variation in T_{eff} . The other columns refer, respectively, to the variations due to $\Delta \log g = +0.3$, $\Delta \xi = +0.4 \text{ km s}^{-1}$, $\Delta[\text{Fe}/\text{H}] = 0.11$, and $\Delta W_{\lambda} = +3.0 \text{ mÅ}$. Column 7 gives the compounded rms uncertainty of the second to sixth columns. The last column gives the observed abundance dispersion of those elements with more than three available lines.

Under the assumption that the errors are independent, they can be combined quadratically so that

$$\sigma = \left(\sum_{i=1}^N \sigma_i^2 \right)^{1/2}.$$

The final abundance uncertainty was found by composing quadratically the uncertainties due to atmospheric parameters and W_{λ} . This final uncertainty is given in Column 7 of Table 5. The last column in this table gives the observed abundance dispersion for the lines of those elements with more than three lines available. These internal uncertainties in the abundances are due to errors in the W_{λ} measurements, in the microturbulence, T_{eff} , and $\log g$ determinations and non-LTE effects which may account for the different lines.

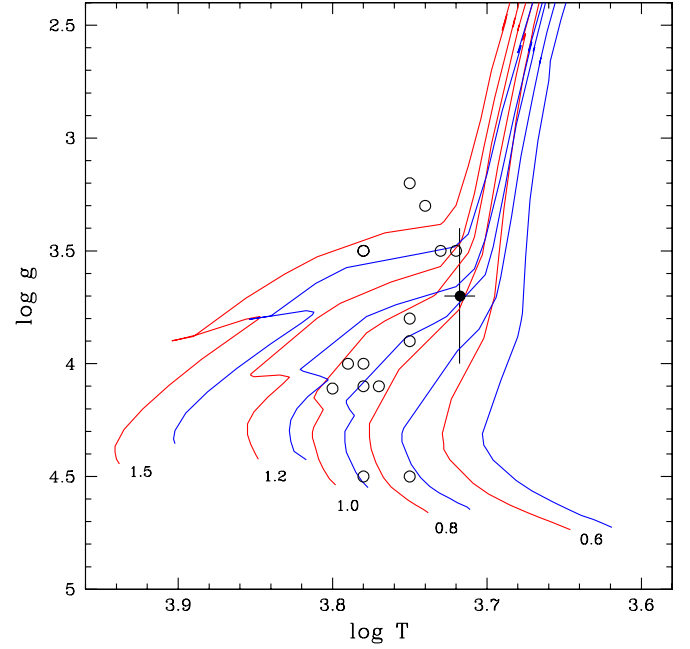


Figure 4. Position of BD-03°3668 (filled circle) and several other CH subgiant stars (open circles) in the $\log g$ – $\log T_{\text{eff}}$ diagram. Evolutionary tracks are from Fagotto et al. (1994). The numbers correspond to stellar masses in units of solar mass. Red lines represent evolutionary tracks for a star with $Z = 0.004$ metallicity and blue with $Z = 0.008$. Data for other CH subgiants were taken from Smith et al. (1993), Luck & Bond (1991), and Tomkin et al. (1989).

Table 6
Sensitivity of the CNO Abundances to the Stellar Parameters Uncertainties and Uncertainties in Carbon and Nitrogen Determinations

Element	ΔT_{eff}	$\Delta \log g$	$\Delta \xi_m$	$\Delta \log(\text{C})$	$\Delta \log(\text{O})$	σ_{tot}
Abundance	+120 K	+0.3	+0.4	+0.1	+0.1	
$\Delta \log \varepsilon(\text{C})$	-0.02	+0.02	0.00	...	+0.04	0.05
$\Delta \log \varepsilon(\text{N})$	+0.13	+0.02	0.00	-0.14	+0.06	0.20
$\Delta \log \varepsilon(\text{O})$ (6300 Å)	-0.01	+0.13	0.00	+0.02	...	0.13

Table 5 shows that neutral elements are rather sensitive to temperature variations, while singly ionized elements are sensitive to variations in $\log g$. For the elements whose abundance determination is based on stronger lines, such as barium, calcium, chromium, and yttrium, the error is introduced also by uncertainties in the microturbulence. The same technique was applied to the elements analyzed via spectrum synthesis, i.e., varying T_{eff} , $\log g$, and ξ , and then computing independently the abundance changes introduced by the variation of the atmospheric parameters. They are also included in Table 5.

We also estimated the influence on the derived CNO abundances of model errors, such as uncertainties in the effective temperature and surface gravity. Besides, abundances of CNO elements are interdependent. Consequently, the uncertainties in the determination of oxygen abundance affect the carbon abundance and vice versa. Uncertainties in carbon abundance result in a variation of nitrogen because the CN molecular lines were used to obtain the N abundance. The variations due to changes in effective temperature (+120 K), surface gravity (+0.3 dex), and C, N, and O abundances (+0.1 dex) are summarized in Table 6. In the last column of this table, we present the resulting abundance uncertainties, σ_{tot} , calculated as the square root of the sum of the squares of the various sources of uncertainty. Derived CNO abundances are weakly sensitive to the

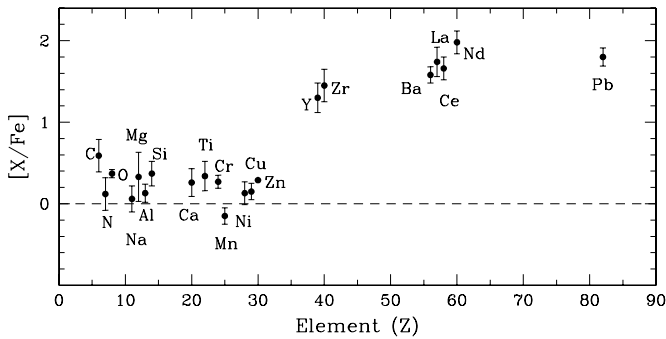


Figure 5. Abundance pattern of BD-03°3668. The error bars were taken from Column 7 of Table 5 and represent the combined errors for each element.

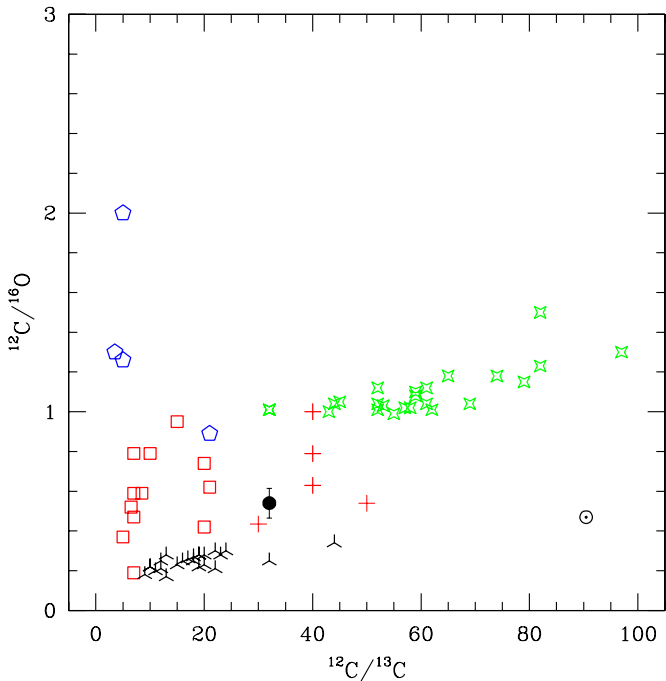


Figure 6. $^{12}\text{C}/^{16}\text{O}$ vs. $^{12}\text{C}/^{13}\text{C}$ ratios for several classes of stars and the star analyzed in this work: barium giants (red open squares), CH stars (blue open polygons), subgiant CH stars (red plus sign), carbon stars (green-shaped point), GK giants (upside down “Y”), and BD-03°3668 (black filled circle). Abundance data for barium giant and dwarf stars are from Smith (1984), Drake & Pereira (2007), and Sneden et al. (1981); CH stars from Vanture (1992); disk carbon stars from Lambert et al. (1986); GK giants from Lambert & Ries (1981).

microturbulent velocity variations since we used weak lines for their determination. A further source of uncertainty in the molecular line analysis comes from the uncertainty in the molecular constants, mainly the dissociation energy of the CN molecule which is not yet well known (Lambert 1994). Our tests showed that variations of the dissociation energy of the CN molecule of $\Delta D_0(\text{CN}) = +0.10$ eV resulted in nitrogen abundance variations $\Delta \log \varepsilon(\text{N}) = -0.10$ dex. Calculations of the carbon isotopic ratio do not depend on the uncertainties in the C and N abundances and in the molecular parameters. The errors in the $^{12}\text{C}/^{13}\text{C}$ determination are mainly due to uncertainties in the observed spectrum, such as possible contamination by unidentified atomic or molecular lines or uncertainties in the continuum placement.

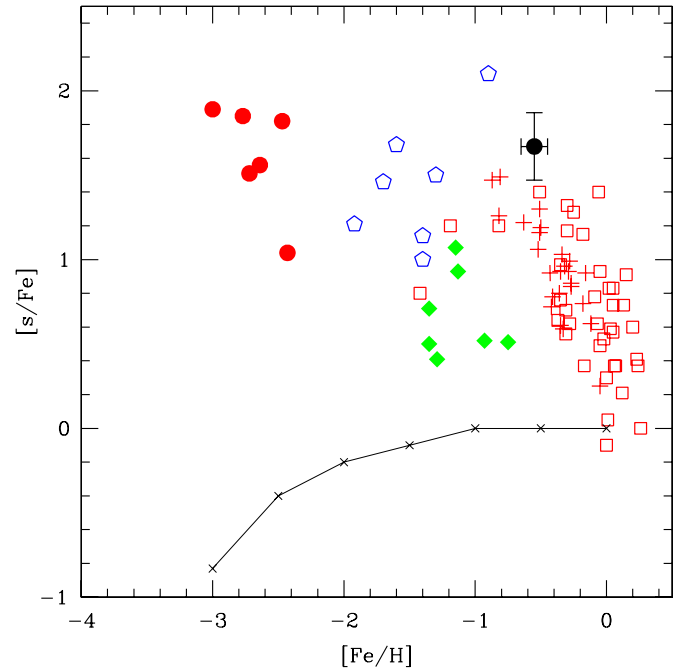


Figure 7. Diagram of $[s/\text{Fe}]$ vs. $[\text{Fe}/\text{H}]$ for several classes of chemically peculiar binary stars. Symbols have the same meaning as in Figure 6. We also show some CEMP stars which are members of binary systems (red filled circles) and yellow symbiotic stars (green filled rhombs). The solid line is the mean $\langle [s/\text{Fe}] \rangle$ for field stars (Gratton & Sneden 1994; Ryan et al. 1996; François et al. 2003).

4. DISCUSSION

4.1. Spectral Type and Distance of BD-03°3668

Previous observations of BD-03°3668 were made by MacConnell et al. (1972), Catchpole et al. (1977), Lu et al. (1983), and Lu (1991) using a prism spectrograph with a dispersion of 50 Å mm^{-1} at $\text{H}\gamma$. All these studies were unable to constrain its luminosity class, and for some stars (MacConnell et al. 1972; Catchpole et al. 1977) not even their spectral types. Lu et al. (1983) provide for BD-03°3668 the spectral-type K1. As a subgiant, at the temperature found in Section 3.2, BD-03°3668 would have a spectral type of around G9-K0 (Schmidt-Kaler 1982).

Since the parallax of BD-03°3668 has not been measured, we estimated the distance of the star using theoretical evolutionary tracks. Figure 4 shows the position of BD-03°3668 in the $\log T_{\text{eff}} - \log g$ diagram, as well as other CH subgiant stars, where model tracks were computed by Fagotto et al. (1994) for metallicities $Z = 0.004$ and 0.008 . Putting the star on the $\log T_{\text{eff}} - \log g$ diagram for stars with metallicity similar to BD-03°3668, we estimated its mass $M_* = (0.8-1.0) M_{\odot}$ and luminosity $L = (3.1-3.8) L_{\odot}$. This corresponds to $M_{\text{bol}} = +(3.5-3.3)$ using $M_{\text{bol}\odot} = +4.74$ (Bessell et al. 1998). Taking into consideration the observed magnitude $V = 10.77$ and $BC = -0.26$ (Alonso et al. 1995) we obtained $M_V = +(3.8-3.6)$ and $r = 250-280$ pc.

4.2. The Abundance Pattern of BD-03°3668

The abundance pattern of BD-03°3668 is shown in Figure 5. Below we will discuss the abundance pattern of this star by comparing it with previous studies carried out on CH subgiants.

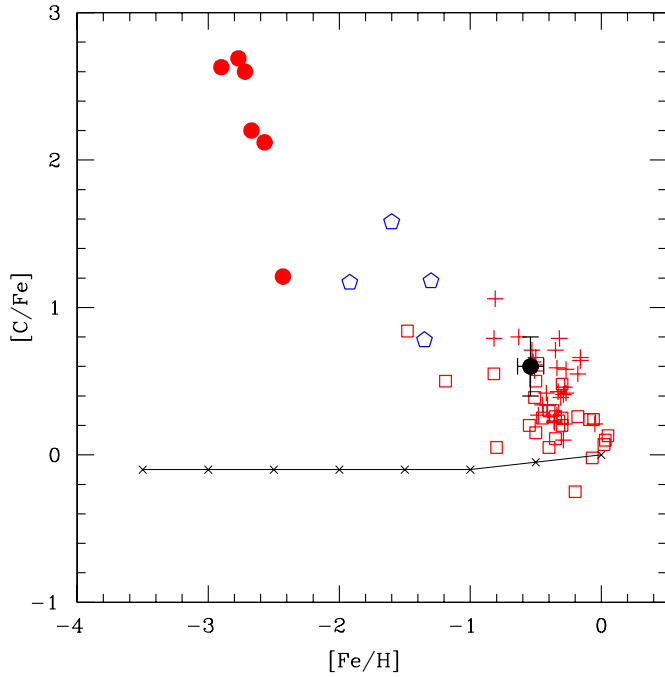


Figure 8. Diagram of $[C/Fe]$ vs. $[Fe/H]$. Symbols have the same meaning as in Figures 6 and 7. The solid line is the mean $\langle [C/Fe] \rangle$ for field stars taken from Masseron et al. (2006).

4.2.1. Nitrogen and Oxygen

Abundance surveys of dwarf stars show that there is no trend for the $[N/Fe]$ ratio versus $[Fe/H]$, i.e., in the metallicity range $-2.0 < [Fe/H] < 0.3$ the $[N/Fe]$ is ≈ 0.0 (Clegg et al. 1981; Tomkin & Lambert 1984; Carbon et al. 1987). Because of the high photospheric temperature ($T_{\text{eff}} \sim 6000$ K) occasionally found in some subgiants, the abundance of nitrogen may become difficult to determine, so only its upper limit can be derived (Allen & Barbuy 2006). The $[N/Fe]$ ratio for BD-03°3668 is similar to the one found in other previous studies of two CH subgiants, HD 50264 and HD 87080, amounting to +0.24 and +0.22 (Drake & Pereira 2007), respectively.

The oxygen abundance follows the same trend as that observed for stars of metallicity similar to BD-03°3668. The mean $[O/Fe]$ for a star with metallicity in the range $-1.0 < [Fe/H] < 0.0$ is approximately 0.3 (Masseron et al. 2006).

4.2.2. $^{12}C/^{13}C$ Ratio

In CH subgiants the $^{12}C/^{13}C$ ratio was investigated by Sneden (1983) and Drake & Pereira (2007). Both studies showed that a high $^{12}C/^{13}C$ ratio is observed in the atmospheres of these stars. Sneden (1983) found that some stars have $^{12}C/^{13}C$ ratio even higher than 40 and Drake & Pereira (2007) obtained, respectively, 30 and 50 for HD 87080 and HD 50264. Compared to giants, this could be taken as an evidence that the first dredge-up has not yet begun in these stars. In fact, in δ Eri, a subgiant analyzed by Lambert & Ries (1981), the $^{12}C/^{13}C$ ratio is higher than 50, while in normal giants the $^{12}C/^{13}C$ ratio is 17.5 ± 0.2 (Lambert & Ries 1981). As seen in Figure 6, CH subgiants are located between the normal GK giants and the carbon stars. Such position suggests that the CH subgiants could have acquired substantial amounts of ^{12}C from an S- or an N-type star with the $^{12}C/^{13}C$ ratio of 20–100 (Smith et al. 1993). According to these authors, the $^{12}C/^{13}C$ ratio observed

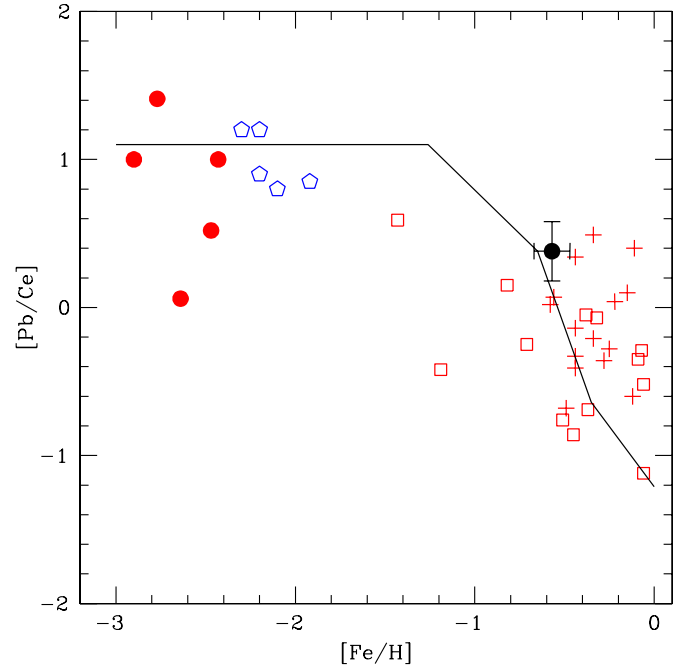


Figure 9. Behavior of the $[Pb/Ce]$ ratio with metallicity. Symbols have the same meaning as in Figures 6 and 7. BD-03°3668 corresponds to the filled black circle located at $[Fe/H] = -0.55$. The solid line represents the prediction from the standard partial mixing (PM) model as given by Goriely & Mowlavi (2000).

in CH subgiants would be consistent with the hypothesis that CH subgiants evolve into barium giants. The derived carbon isotopic ratio for BD-03°3668 is $^{12}C/^{13}C \geq 32$, in agreement with the previous determinations for this kind of star.

4.2.3. Li

Lithium abundance was estimated by matching the observed spectrum of BD-03°3668 in the region of the Li I $\lambda 6708$ resonance doublet with the synthetic spectra. The wavelengths and oscillator strengths for the individual hyperfine and isotopic components of the lithium lines were taken from Hobbs et al. (1999). A solar $^6\text{Li}/^7\text{Li}$ isotopic ratio (0.081) was adopted to calculate the synthetic spectrum. The list of CN lines in the vicinity of the Li I doublet was taken from Mandell et al. (2004) who improved the theoretical positions and oscillator strengths of the CN lines by comparison to a carbon arc spectrum. In our synthesis, we modified the CN line positions taken from Mandell et al. (2004) by -0.03 Å in order to achieve a better fit to the observed spectrum of BD-03°3668. The atomic line list is essentially the same as described in Drake & Pereira (2008) and Pereira & Drake (2009). The line $\lambda 6708.099$ of Ce II is well fitted to the obtained cerium abundance. However, the line at 6707.740 Å found in the spectra of the barium stars by Lambert et al. (1993) and supposed to be a Ce II line could not be fitted with the oscillator strength and the excitation potential suggested by these authors. We determined the upper limit of the lithium abundance to be $\log \varepsilon(\text{Li}) \leq -0.1$.

4.2.4. Other Elements: Na to Zn

The mean abundance of α -elements (Mg, Si, Ca, and Ti) of BD-03°3668 is 0.32 ± 0.10 . The mean $[\alpha/Fe]$ ratio given by $[\langle \text{Mg, Si, Ca} \rangle / 3Fe]$ for a sample of thick disk F and G dwarfs lies between 0.2 and 0.3 for metallicities $-0.5 < [Fe/H] < -0.3$ (Reddy et al. 2006). The $[\alpha/Fe]$ ratio for BD-03°3668 is

slightly higher than that found in other CH subgiants: 0.14 ± 0.17 for a sample of subgiant CH stars analyzed in Smith et al. (1993, hereafter SCL93) (taking into consideration only silicon and calcium); 0.18 ± 0.13 (Luck & Bond 1991), 0.12 ± 0.03 (Pereira 2005); 0.11 (Pereira & Junqueira 2003), $0.2 - 0.3$ (Allen & Barbuy 2006).

The abundances of the iron group elements follow the Fe abundance. Their mean [element/Fe] ratio, including nickel, chromium, zinc, copper, and manganese is 0.14 ± 0.11 similar to other determinations for CH subgiant stars, 0.06 (SCL93), 0.08 (Luck & Bond 1991), 0.00 (Allen & Barbuy 2006), 0.04 (Pereira 2005), and 0.09 (Pereira & Junqueira 2003).

Sodium shows the same trend as the thick disk dwarf stars with a mean [Na/Fe] ratio of ~ 0.1 for a metallicity between -0.3 and -1.0 (Reddy et al. 2006). Aluminum is an odd-Z element and should be expected to behave like Na, another odd-Z element. For a metallicity around -0.5 , the typical [Al/Fe] ratio is $+0.2$ with a large scatter (Reddy et al. 2006).

4.2.5. Carbon and *s*-process Elements

Figure 5 shows that a general overabundance of elements heavier than iron synthesized by neutron capture is observed in BD-03°3668. The available abundances for low-metallicity thin/thick disk dwarf stars (Reddy et al. 2003, 2006) indicate that the chemical evolution of the Galaxy could not be responsible for the *s*-process element overabundances observed in BD-03°3668. The abundance pattern of BD-03°3668 implies a high abundance of the *s*-process elements, $[s/\text{Fe}]$ of $+1.63$, where “*s*” represents the mean abundance of the elements created by slow neutron capture reactions (*s*-process): Y, Zr, Ba, La, Ce, Nd, and Pb. Similar values were found in other studies of CH subgiants (Pereira & Junqueira 2003; Pereira 2005; Allen & Barbuy 2006). The $[s/\text{Fe}]$ versus $[\text{Fe}/\text{H}]$ relation for a sample of barium giants and CH subgiants, as well as for BD-03°3668 analyzed in this work, is shown in Figure 7. The position of BD-03°3668 in this figure is the same as for other stars of the same class. As Clayton (1988) showed, a clear, significant trend of increasing $[\text{hs}/\text{ls}]$ (as well as $[s/\text{Fe}]$) with decreasing metallicity is the result of the operation of the reaction $^{13}\text{C}(\alpha, n)^{16}\text{O}$ as a neutron source. Similar to the $[s/\text{Fe}]$ index, the carbon abundance is also anticorrelated with metallicity. Figure 8 shows the $[\text{C}/\text{Fe}]$ abundance ratio plotted as a function of the metallicity given by $[\text{Fe}/\text{H}]$ for the same objects as in Figure 7.

Figure 9 shows the $[\text{Pb}/\text{Ce}]$ ratio as a function of metallicity. In this figure, we show BD-03°3668 (the filled black circle at $[\text{Fe}/\text{H}] = -0.55$), the CH stars analyzed by Van Eck et al. (2003), CEMP stars which are members of binary systems, and the barium giants and dwarfs from Allen & Barbuy (2006). The position of BD-03°3668 in this diagram follows the same trend as the other barium dwarfs and giants investigated so far.

5. CONCLUSIONS

An abundance analysis, employing a high-resolution optical spectrum, showed that BD-03°3668 is a new CH subgiant substantially enriched with heavy *s*-process elements. The abundance pattern of the iron peak elements, of the elements of the α group, as well as that of sodium and aluminum follows the disk abundance. The derived metallicity of BD-03°3668 ($[\text{Fe}/\text{H}] = -0.55$) shows that this star is another metal-poor CH subgiant just like the others analyzed so far. CH subgiant stars with metallicities close to solar are still to be found (Začs 1994; Pereira 2005).

This research used the SIMBAD database, operated at CDS, Strasbourg, France.

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